

ASSESSMENT TOOL - 1

Course code	CSA0702	Course Title	computer networks
Co Assessed	CO1	Assessment	Total - short Answers
weightage		Total Marks	20 - Marks

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Section Batch	AIBDS - 2024	Date of submission	17 / July / 2026
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Total Marks	20 M	obtained Marks	20
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understanding of co-cepts	explanation	Application	organization	Accuracy	clarity
25%	25%	20%	15%	10%	5%
5	5	4	3	2	1

Step 1 :- payload per fragment
 $1500 - 20 = 1480 \text{ bytes}$

Step 2 :- Number of fragments
 $\frac{12000}{1480} = 8.1$

Round up

Fragments = 9

Step 3 :- Header overhead

Each fragment has one 20 byte header.

$$9 \times 20 = 180 \text{ bytes}$$

Total header overhead = 180 bytes

Explanation :-

The Network layer fragments packets when they exceed the MTU of a network.

Q4. Sliding window protocol

Given

- Window size = 10 frames
- Data frame = 1024 frames bytes
- Total frames = 48

Step 1 :- Number of transmission windows

$$\frac{48}{10} = 4$$

Transmission windows = 8

Step 2 :- Retransmissions

One frame is lost in every window

$$10 \times 1 = 10$$

Total retransmissions = 4 frames

Explanation:-

The transport layer uses the sliding window protocol for flow control

Qs. Session layer synchronization.

Given

- Video file = 600 MB
- Checkpoint every = 60 MB
- Failure after = 390 MB

Step 1: - Number of checkpoints before failure

Completed checkpoints

60, 120, 180, 240, 300, 360

Number of checkpoints = 6

Step 2: Data to retransmit

The last checkpoint is at 360 MB

Failure occurs at 390 MB

Data transmitted after the last checkpoint:

$$390 - 360 = 30 \text{ MB}$$

Data to retransmit = 30 MB

Explanation:-

The session layer manages communication sessions and inserts synchronization checkpoints.

Presentation Layer compression and Encryption.

Given

- Original file = 900 MB
- compression = 40%
- Encryption overhead = 5%

Step 1: compressed file size

40% reduction means 60% remains

$$900 \times 0.60 = 540 \text{ MB}$$

compressed size = 540 MB

Step 2:- Encrypted file size

5% of 540 MB

$$540 \times 0.05 = 27 \text{ MB}$$

Encrypted size

$$540 + 27 = 567 \text{ MB}$$

encrypted file size = 567 MB

Step 3: overall reduction

Reduction

$$900 - 567 = 333 \text{ MB}$$

percentage reduction

$$\frac{333}{900} \times 100 = 37\%$$

Total reduction = 37%

Explanation:-

The presentation layer performs data compression to reduce file size, saving bandwidth

Q7. FTP Application Data Transfer

Given

- file size = 3 GB
- Throughput = 120 Mbps
- Request size = 3 MB

Step 1: Convert GB to MB

$$3 \text{ GB} = 3000 \text{ MB}$$

Step 2: - Number of requests

$$\frac{3000}{3} = 1000$$

Total requests = 1000

Step 3: - Transmission time

Convert data to megabits

$$3000 \text{ MB} \times 8 = 24000 \text{ Mb}$$

$$\frac{24000}{120} = 200 \text{ seconds}$$

Transmission time = 200 seconds

Explanation: -

The Application layer provides services such as FTP for reliable file transfer.

Q8. End to-End Delay.

Given

- Application delay = 4ms
- Transport delay = 3ms
- Network delay = 5ms

Data Link delay = 2ms

Physical delay = 1ms

Propagation delay = 18ms

Transmission delay = 12ms

Step 1: processing delay

$$4 + 3 + 5 + 2 + 1 = 15ms$$

Step 2: Total delay

$$15 + 18 + 12 = 45ms$$

Explanation:

Each OSI layer performs a specific function before forwarding data.

Q9. protocol overhead and Efficiency.

Given

useful data = 80MB

Frame size = 1600 bytes

Overhead / frame = 80 bytes.

Step 1: payload per frame

$$1600 - 80 = 1520 \text{ bytes}$$

Step 2: useful data in bytes

$$80 \times 1,000,000 = 80,000,000 \text{ bytes}$$

Step 3: Number of frames

$$\frac{80,000,000}{1520} = 52,632$$

Step 4: Total overhead

$$52,632 \times 80 = 4,210,560 \text{ bytes}$$

Steps Transmission efficiency

$$\frac{1520}{1600} \times 100 = 95\%$$

Explanation:-

Protocol overhead consists of headers and trailers added by different OSI layers.

Q10 Health care Network Data Transfer.

Given

Original file = 240 MB

Compression = 30 %.

Segment size = 1400 bytes

Data Link overhead = 60 bytes / frame.

Step 1: compressed file size

70% of original

Compressed size = 168 MB

Step 2: convert to bytes

$$168 \times 1,000,000 = 168,000,000 \text{ bytes}$$

Step 3: Number of segments

$$\frac{168,000,000}{1400} = 120,000$$

Step 4: Total protocol overhead

$$120,000 \times 60 = 7,200,000 \text{ bytes}$$

Step 5: final data transmitted

$$168 + 7.2 = 175.2 \text{ MB}$$

Explanation:-

Multiple OSI layers work together during transmissions